

Lab Experiment 1

Title

- i. Installing Linux Operating System
 - ii. Boot, Reboot, and Shut Down a System Normally
 - iii. Boot Systems into Different Run Levels Manually
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Aim

1. To install the Linux Operating System using dual boot or a virtual machine.
 2. To understand the process of booting, rebooting, and shutting down a system.
 3. To manually boot the system into different run levels (or systemd targets).
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Requirements

- A computer with minimum 4 GB RAM and 40 GB free storage.
 - Linux ISO image (Ubuntu recommended).
 - VirtualBox/VMware software (for Virtual Machine installation).
 - Administrative privileges to access BIOS/UEFI settings.
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Theory

1. Linux Installation Options

Option A: Dual Boot Installation

- In dual boot, Linux is installed **alongside another OS (e.g., Windows)**.
- On startup, the user chooses which OS to boot using the **GRUB bootloader**.
- Steps:
 1. Create a bootable USB with the Linux ISO.
 2. Boot from USB and select *Install Ubuntu*.
 3. Choose *Install alongside Windows* (or *Something Else* for manual partitioning).
 4. Create partitions for root (/), swap, and home (/home).

5. Proceed with installation and restart the system.

Option B: Virtual Machine Installation (Ubuntu on VirtualBox)

- Virtual Machines allow installing Linux inside another OS without modifying partitions.
 - Steps:
 1. Install **Oracle VirtualBox** on your system.
 2. Open VirtualBox → *New* → Name: *Ubuntu* → Select Type: Linux, Version: Ubuntu.
 3. Allocate memory (min 2 GB).
 4. Create a virtual hard disk (20 GB or more, VDI, dynamically allocated).
 5. Start VM → Choose Ubuntu ISO as boot media.
 6. Begin Ubuntu installation:
 - Select language and keyboard layout.
 - Choose *Normal Installation*.
 - Select disk partitioning (Erase disk → Use entire virtual disk).
 - Create username and password.
 - Complete installation and restart.
 7. Ubuntu boots into a fresh installation inside VirtualBox.
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2. Boot, Reboot, and Shut Down

- **Booting Process in Linux:**
 1. **BIOS/UEFI** – Initializes hardware.
 2. **GRUB Bootloader** – Loads Linux kernel.
 3. **Kernel** – Initializes devices and mounts root filesystem.
 4. **Init/Systemd** – Starts processes and services.
 5. **Login Screen** – CLI or GUI depending on runlevel/target.
- **Commands for Reboot and Shutdown:**
 - `reboot` # Reboot the system
 - `shutdown -r now` # Restart immediately

- `shutdown -h now` # Shutdown immediately
 - `poweroff` # Power off the system
 - `halt` # Halt the system
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3. Boot Systems into Different Run Levels Manually

1. What are Run Levels? (Old SysV systems)

- In Linux, a runlevel defines the mode of operation of the system.
- It tells the OS what services/processes should be running after boot.
- Example: A server may run without GUI (text-only), while a desktop may boot into GUI.

Standard Runlevels (SysV Init):

Runlevel	Meaning
0	Halt (system shutdown)
1	Single-user mode (for maintenance, no networking, root only)
2	Multi-user mode (without network, varies by distro)
3	Multi-user mode (with networking, CLI only)
4	Undefined (custom use)
5	Multi-user mode with GUI (graphical desktop)
6	Reboot

2. What are Systemd Targets? (Modern Systems)

- Newer Linux uses systemd instead of SysV init.
- Runlevels are replaced with targets (but concept is same).

Systemd Targets:

Runlevel (old)	systemd Target	Meaning
0	poweroff.target	Shutdown
1	rescue.target	Single-user mode
3	multi-user.target	Multi-user (CLI)
5	graphical.target	Multi-user (GUI)
6	reboot.target	Reboot

3. Why Do We Need Different Run Levels?

- **Maintenance:** Boot into single-user mode if system is broken.
 - **Servers:** Often run in multi-user mode (no GUI) to save resources.
 - **Desktops:** Normally boot in graphical mode.
 - **Customization:** Admin can choose default runlevel/target.
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4. How to Boot Manually into Different Run Levels?

Step A: Check Current Runlevel/Target

- **For SysV:**
-> runlevel

Output example:

N 5 # currently in runlevel 5 (graphical)

- **For Systemd:**
-> systemctl get-default

Output example:

graphical.target

Step B: Switch Runlevels Temporarily

- SysV Init (older distros like RHEL 5/6):

`init 3` # Switch to multi-user CLI

`init 5` # Switch to GUI

`init 1` # Single-user mode

- Systemd (modern Ubuntu, Fedora, RHEL 7+):

`systemctl isolate multi-user.target` # Switch to CLI mode

`systemctl isolate graphical.target` # Switch to GUI mode

`systemctl isolate rescue.target` # Single-user mode

👉 These changes are temporary — once rebooted, the system goes back to default.

Step C: Set Default Runlevel/Target (Permanent)

- SysV Init:

- Edit `/etc/inittab` and set default runlevel. Example:

- `id:3:initdefault:` # Boots in CLI mode

- `id:5:initdefault:` # Boots in GUI mode

- Systemd:

`systemctl set-default multi-user.target`

`systemctl set-default graphical.target`

6. Key Points

- Runlevels = modes of Linux.
- Old systems (SysV) use `init`, modern systems (Systemd) use `systemctl`.
- Temporary change: `init <n>` or `systemctl isolate <target>`.
- Permanent change: `/etc/inittab` (old) or `systemctl set-default` (new).
- Single-user mode is useful for fixing system problems.
- Multi-user CLI is common for servers.
- Graphical mode is common for desktops.

Procedure

1. Install Linux OS

- Choose either dual boot or VirtualBox installation.
- Complete installation and log into the system.

2. Boot, Reboot, and Shutdown

- Use the reboot, shutdown, halt, and poweroff commands.
- Observe system behavior after each.

3. Runlevels and Targets

- Check the default target using `systemctl get-default`.
- Switch between multi-user mode (CLI) and graphical mode.
- Experiment with rescue mode to see minimal system state.

Observations

- The Linux OS successfully installs and boots.
- Reboot and shutdown commands function as expected.
- Switching between runlevels/targets changes the system mode (CLI, GUI, rescue).

Result

The Linux operating system was successfully installed. The system was booted, rebooted, and shut down using appropriate commands. Students also learned to manually switch the system into different runlevels/targets and understood their significance.
